

ing its cause. It fails to answer the question of what it is that computers do—or what it is that people do with computers—that causes educated workers to be relatively more in demand.

This paper proposes an answer to this question. We formalize and test a simple theory of how the rapid adoption of computer technology—spurred by precipitous real price declines—changes the tasks performed by workers at their jobs and ultimately the demand for human skills. Our approach builds on an intuitive set of observations offered by organizational theorists, computer scientists, and most recently economists about what computers do—that is, the tasks they are best suited to accomplish—and how these capabilities complement or substitute for human skills in workplace settings.² The simple observations that undergird our analysis are (1) that computer capital substitutes for workers in carrying out a limited and well-defined set of cognitive and manual activities, those that can be accomplished by following explicit rules (what we term “routine tasks”); and (2) that computer capital complements workers in carrying out problem-solving and complex communication activities (“nonroutine” tasks). (See Table I for examples.) Provided that routine and nonroutine tasks are imperfect substitutes, these observations imply measurable changes in the task composition of jobs, which we test below.

To answer the core questions of our paper, the ideal experiment would provide two identical autarkic economies, one facing a dramatic decline in the price of computing power and the other not. By contrasting these economies, it would be straightforward to assess how computerization reshapes the task composition of work and hence the structure of labor demand. Because this experiment is not available, we develop a simple economic model to predict how demand for workplace tasks responds to an economywide decline in the price of computer capital. The model predicts that industries and occupations that are initially intensive in labor input of routine tasks will make relatively larger investments in computer capital as its price declines. These industries and occupations will reduce labor input of routine tasks, for which computer

2. Simon [1960] provides the first treatment of this question with which we are familiar, and his essay introduces many of the ideas explored here. Other early works include Drucker [1954] and Nelson and Winter [1982]. Adler [1986], Orr [1996], and Zuboff [1988] discuss what computers and related technology do in the workplace, but do not consider economic implications. Acemoglu [1998], Goldin and Katz [1998], Bresnahan [1999], Bartel, Ichniowski, and Shaw [2000], Lindbeck and Snower [2000], Lang [2002], and Bresnahan, Brynjolfsson, and Hitt [2002] provide economic analyses of why technology and human capital are complementary.

bution—college graduates and high school dropouts—similar patterns prevail, but they are less precisely estimated. In all cases, the estimates are of the expected sign, but none is statistically significant. For college graduates, this is likely to reflect “topping out,” since this education group was already at the extreme of the distribution for all tasks. We are less certain why the relationships are weaker for high school dropouts, but one possibility is that this group has insufficient human capital to be effectively redeployed to alternative job tasks.

To assess whether these within-education group shifts are a quantitatively important component of the overall change in industry task content, panel F of Table V presents a decomposition of industry task changes into within and between education group components. This exercise shows that in every case, within-education group task upgrading explains a substantial share, 24 to 111 percent, of total task upgrading over these two decades. For example, the annual within-industry change in nonroutine interactive tasks over the period 1980–1998 is 4.4 centiles per decade, of which 3.1 centiles (71 percent) is accounted for by contemporaneous industry computerization. Within-education group task changes explain the bulk of these shifts: 78 percent of the explained component and 55 percent of the total. Subdividing the explained within-education group component further, 59 percent is due to changes in task assignment among high school graduates and those with some college, and the rest is equally accounted for by task shifts among college graduates and high school dropouts.

This exercise demonstrates that within-education group shifts in task content are the primary channel through which the structure of workplace tasks has shifted over the past two decades. Furthermore, a large portion of the within-education group changes are accounted for by cross-industry patterns of computer adoption. This suggests to us that task change is antecedent to educational upgrading, rather than merely a reflection of it.

V.B. Task Shifts within Occupations

The analyses above exploit shifts in occupational composition—the extensive margin—to quantify changes in task input. This approach is imperfect since it assumes that the tasks performed within occupations are static, which is unlikely to be accurate over long time intervals. Moreover, our task framework implies that this assumption should be violated in a specific

manner: occupations undergoing rapid computerization should differentially reduce labor input of routine cognitive and manual tasks and increase labor input of nonroutine cognitive tasks. To provide one example, the 1976 edition of the Department of Labor's *Occupation Outlook Handbook* described the job of Secretary as: "... Secretaries relieve their employers of routine duties so they can work on more important matters. Although most secretaries type, take shorthand, and deal with callers, the time spent on these duties varies in different types of organizations" [U. S. Department of Labor 1976, p. 94]. In 2000 the entry for Secretary reads: "As technology continues to expand in offices across the Nation, the role of the secretary has greatly evolved. Office automation and organizational restructuring have led secretaries to assume a wide range of new responsibilities once reserved for managerial and professional staff. Many secretaries now provide training and orientation to new staff, conduct research on the Internet, and learn to operate new office technologies" [U. S. Department of Labor 2000, p. 324].

To test whether this example captures a pervasive phenomenon, we match occupations from the 1977 and 1991 revisions of the DOT to estimate the following equation:

$$(16) \quad \Delta T_{mk\tau} = \alpha + \xi \Delta C_m + \varepsilon_{mk\tau}.$$

Here, $\Delta T_{mk\tau}$ is the change in occupational input of task k between 1977 and 1991 in three-digit COC occupation m , and ΔC_m is the change in occupational computer penetration measured by the CPS. To provide a clean test, our data set is constructed using only the subset of occupations appearing in the 1977 DOT, which was used to create our original occupation crosswalk. Accordingly, the variation used to estimate equation (16) stems exclusively from DOT examiners' reevaluation of the task content of individual occupations between 1977 and 1991.²³

Table VI presents three estimates for each task measure. The first column of each panel presents a bivariate regression of the within-occupation change in task content on occupational computerization and a constant. These estimates provide striking

23. The weighted fraction of employment reevaluated between 1978 and 1990 in our data is 73 percent. Occupations were chosen for reevaluation by DOT examiners partly on the expectation that their content had changed. Hence, this is not a random sample. We assume that occupations that were not revised between the 1977 and 1991 DOT experienced no task change. Provided that these occupations did not experience offsetting shifts, our approach will provide a lower bound on the extent of task change.